

One interval k , one state polynomial

whole interval: $h_k = 2$

$$p_k(\tau) = 1 + \tau + 2\tau^2$$

one piece across the entire interval

$$x_{k+1} = p_k(1) = 4$$

$$x_k = 1$$

$$x_{k,i} = p_k(\tau_i) = 2$$

stage i is this point

local τ

0

$$\tau_i = 1/2$$

1

time t

$$t_k = 2$$

$$t_k + h_k \tau_i = 3$$

$$t_{k+1} = 4$$



integrate to stage i

$$x_{k,i} = x_k + h_k \sum_j A_{ij} f_{k,j} = 1 + 1 = 2$$

Row i of A integrates from 0 to τ_i .



integrate to the right endpoint

$$x_{k+1} = x_k + h_k \sum_j b_j f_{k,j} = 1 + 3 = 4$$

The weights b integrate from 0 to 1.